

THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

CCST5037: Mathematics: A Cultural Heritage
Guidelines on Group Project (Essay & Presentation)

- (1) Work in groups of either two or three. Project-group members must be in the same tutorial group, because your project-group is to give its oral presentation in one of the tutorial sessions.
- (2) The length of the essay must be in the range of **2000 – 3000** words, excluding figures and bibliography. It is on a book (or a piece of work, e.g., a movie) of your choice about mathematics that is relevant to the learning outcomes of the course. If you like, your essay can be on more than one book/piece of work, as long as they are related; or, your essay can be on a part of the book instead of the whole book, as long as your essay is rich in content and complete in itself. It should not be a simple book report; you must also write your interpretation and the relevance to the theme(s) of this course.
- (3) **Essay plan (assignment 1):** To ensure that you are on the right track, each group needs to submit their **essay plan** before **16:55, October 16th, 2026 (Friday)**. Along with your initial score out of 10 points, feedback will then be given, and the score for your essay plan may be increased by resubmission of an updated project plan, due **16:55, October 29th, 2026 (Thursday)**. Only one representative per group needs to submit the plan on behalf of the whole group. Your essay plan should include: (i) tentative title of your essay, together with an explanation of what drew your group to this topic, (ii) names and UID of group members and tutorial group, (iii) the book(s) (or piece of work) you are going to write about, and (iv) a proposed central theme (v) a brief description of the proposed essay content, how this is related to the course learning outcomes. You will receive feedback that should help guide you in writing the final essay.
- (4) **Oral presentation:** Your group will give an oral presentation of your project (on the same topic as your essay) during the tutorial sessions starting from **November 2nd, 2026**. Upload your **presentation file** (in ppt, pptx, or pdf format) to Moodle before **16:55, November 1st, 2026 (Sunday)**. Late penalties (up to 10 out of 15 points for the presentation) will be assessed as follows:
 - 1) 0.2 points for submission within the first 3 minutes after the deadline, then
 - 2) 0.5 points for submission up to 10 minutes after the deadline, then
 - 3) 1 point for submission within the first 30 minutes after the deadline, then
 - 4) 1 points for every 30 minute period after that, up to 4 points per day (so 2 hours-24 hours late would be a 4 point penalty)
 - 5) On day 2: 1 point for every 30 minutes, up to 4 points per day (so 26-48 hours late would be an 8-point penalty)
 - 6) On day 3, 1 point for every 30 minutes, up to 2 points (giving a 10-point penalty after 2 days, 1 hour).

Length of each presentation is 12 minutes, followed by a 3-minute Q&A session. All presentations must be done on time; in case any group members are late, he or she will receive a penalty of 5 marks (out of 15) for the presentation part.

- (5) **Due date of your group essay is 16:55, December 1st, 2026 (Tuesday)**. Only one member of your group should submit the final project, but you are still responsible for submission, so make sure that a group member has been designated to submit. Make sure to submit early to avoid technical problems that may arise with the submission; late submission will not be graded.

More on the books (or works) that you may use as basis of your essay

The book you select should **not** be a book of technical content on a certain subject in mathematics per se. For instance, do not select a book with a title like *Introduction to Number Theory* or *A First Course in Calculus*. Specifically, you are permitted to include technical aspects in your essay, but your concentration should be on the impact and interactions with society via the Course Learning Outcomes, not the technical aspects themselves.

A possible selection includes a popular account on mathematical topics, an exposition on the nature or on the philosophical aspect of mathematics, a novel or a play with a mathematical flavour, biographies of mathematicians, etc. There are plenty of suitable books. The level may vary with the background of mathematical knowledge and mathematical maturity of the readership the author has in mind. In this course we do not require advanced level mathematics, but if you wish to choose a book that demands more on your mathematical knowledge, you are welcome to do so. A **small warning** for those who choose more technical topics for their group project: **remember your audience** and also the **Course Learning Outcomes**; if you choose a highly technical topic, you should remember that you must explain what you are investigating to the other students in your tutorial (you are not merely giving a presentation to the teachers of the course) and you also need to explain how this technical aspect is **important within the context of society and culture** or in the **“way” of mathematics**, per the goals of the course. Every semester there is a very nice presentation given of interesting mathematics by at least one group, but this has not always necessarily been a successful presentation, as some instances did not bring these concepts into the context investigated in this course.

The list below is not to be regarded as a recommended booklist. Rather, treat it as a list of examples. By browsing through some of them you should get some idea about what would be suitable, and you may select any suitable book which is not necessarily on this list. Most of the books in the list are available in the University Library, with their call numbers shown, but you need not restrict your choice to the holding in the University Library.

Development / History of mathematics

- J. McLeish, *The Story of Numbers: How Mathematics Has Shaped Civilization*, Fawcett Columbine, 1994 [510.9M16]
- W. Dunham, *Journey through Genius: The Great Theorems of Mathematics*, Wiley, 1990. [510.9 D91]
- H.W. Eves, Howard Whitley, *An introduction to the history of mathematics (with cultural connections by Jamie H. Eves)*, 6th ed., Saunders College Pub., 1990. [510.9 E93]
- G.G. Joseph, *The Crest of the Peacock: Non-European Roots of Mathematics*, Princeton University Press, 2011. [510.9 J832 c92]

General interest

- I. Stewart, *From Here to Infinity: A Guide to Today's Mathematics*, Oxford University Press, 1996 [510 S84 f ; electronic version available]
- D.R Hofstadter, *Gödel, Escher, Bach: An Eternal Golden Braid*, Basic Books, 1999. [164 H71]
- J.P. King, *The Art of Mathematics*, Plenum Press, 1992.[510 K53]
- C. Henrion, *Women in Mathematics: the addition of difference*, netLibrary, Inc., 2000 [electronic version available from HKU Library]
- T. Gowers, *Mathematics: a Very Short Introduction*, Oxford University Press, 2002.[510 G7]

- S. Stein, *How the Other Half Thinks*, McGraw Hill, 2001. [510 S819 h; electronic version available from HKU Library]
- P.J. Davis, R. Hersh, *The Mathematical Experience*, Study Edition (with E.A. Marchisotto), Birkhäuser, 1995. [510.1 D26]
- P.J. Davis, *Mathematics and Common Sense: A Case of Creative Tension*, A.K. Peters, 2006 [511.3 D2]
- J.A. Paulos, *Innumeracy: Mathematical Illiteracy and Its Consequences*, Vintage Books, 1990. [510 P33i]
- J.A. Paulos, *Mathematics and Humor*, University of Chicago Press, 1982, [808.7 P33]
- A. Bellos, *Alex's Adventures in Numberland*, Bloomsbury, 2010. [B 510 B447]
- I. Stewart, *Why Beauty Is Truth: A History of Symmetry*, Basic Books, 2007. [B 539.725 S85]
- M.R. Huber, *Mythematics: Solving the Twelve Labors of Hercules*, Princeton University Press, c2009. [510 H877 m99]
- I. Stewart, *The Problems of Mathematics*, Oxford University Press, 1992. [X 510 S84 p]
- K.J. Devlin, *Life By the Numbers*, Wiley, 1998. [510 D497]; electronic version available from HKU Library]
- S.K. Stein, *Strength in Numbers: Discovering the Joy and Power of Mathematics in Everyday Life*, Wiley, 1996. [510 S819V]

Biographies

- P. Hoffman, *The Man Who Loved Only Numbers: The Story of Paul Erdős and the Search for Mathematical Truth*, Hyperion, 1998.
- L.M. Osen, *Women in Mathematics*, MIT Press, 1974 [X 510A O8; electronic version available from HKU Library]
- R. Kanigel, *The Man Who Knew Infinity: A Life of the Genius Ramanujan*, C. Scribner's, 1991. [510B R165K]
- S. Nasar, *A Beautiful Mind*, Faber and Faber, 1998. [510B N25 N]
- G.H. Hardy, *A Mathematician's Apology*, reprinted with foreword by C.P. Snow, Cambridge University Press, 1992. [510 H269 m; electronic version available from HKU Library]
- H. Wang, *Reflections on Kurt Gödel*, MIT Press, 1987. [164B G58 W; electronic version available from HKU Library]
- P.R. Halmos, *I Want To Be A Mathematician: an Automathography*, Springer, 1985. [510B H19]
- Y. Wang, Hua Loo-Keng, a Biography (translated into English by P. Shiu), Springer, 1999. [510B H87 W]

Topics in mathematics

- P. Beckmann, *A History of Pi*, Golem Press, 1977. [512.924 B39]
- S. Singh, *Fermat's Last Theorem: The Story of a Riddle That Confounded the World's Greatest Minds for 358 years*, Fourth Estate, 1997. [512.74 S61]
- A. Aczel, *Fermat's Last Theorem: unlocking the secret of an ancient mathematical problem*, Four Walls Eight Windows, 1996. [512.74 A18]
- P.J. Nahin, *An Imaginary Tale: The Story of $\sqrt{-1}$* , Princeton University Press, 1998. [515.9 N153 i]
- E. Maor, *e: The Story of a Number*, Princeton University Press, 2009. [512.73 M29]

Mathematics and cognitive science

- S. Dehaene, *The Number Sense: How The Mind Creates Mathematics*, Oxford University Press, 2011. [510.19 D322 n97]
- G. Lakoff and R. Núñez, *Where Mathematics Comes From: How The Embodied Mind Brings Mathematics Into Being*, Basic Books, 2000. [510 L192 w]

Mathematics and philosophy

- R. Hersh, *What is Mathematics, Really?* Jonathan Cape, 1997. [510.1 H57; electronic version available from HKU Library]
- R.v.B. Rucker, *Infinity and The Mind: The Science and Philosophy of The Infinite*, Princeton University Press, 2005. [511.3 R91]

Novels / Stories / Plays

- E.A. Abbott, *Flatland: A Romance of Many Dimensions* (2nd revised edition, 1884), Cambridge University Press, 2010. [530.11 A131 f58; electronic version available from HKU Library]
- S.H. Strogatz, *The Calculus of Friendship: What a Teacher And a Student Learned About Life While Corresponding About Math*, Princeton University Press, 2009. [510.922 S921 c14]
- A. Doxiadis, *Uncle Petros and Goldbach's Conjecture: A Novel of Mathematical Obsession*, Bloomsbury USA, 2001.
- A.P. Lightman, *Good Benito*, Pantheon Books, 1994 [815.1 L72 C3 g]
- Y. Ogawa, *The housekeeper and the professor* (translated by Stephen Snyder), Picador, 2009. [895.641 O34 C3 h] (Also in feature film)
- D. Auburn, *Proof: A Play*, Faber and Faber, 2001. [815.1 A897 C2 p] (Also in feature film)

Movies

- *Proof*, 2005.
- *A Beautiful Mind*, 2001.
- *Good Will Hunting*, 1997.
- *The housekeeper and the professor / The Professor and His Beloved Equation* (Japanese movie), 2006.
- *The Imitation Game*, 2014

Comics / Graphic Novel

- A. Doxiadis, C.H. Papadimitriou, A. Papadatos, *A Di Donna, Logicomix: An Epic Search for Truth*, Bloomsbury USA, 2009.